

Binh Minh Tran

Honours Student, Computational Data Science, School of Computer Science, The University of Sydney

CONTACT INFORMATION

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RESEARCH PROFILE

Final-year Bachelor of Advanced Computing (Honours) candidate at the University of Sydney (WAM 82/100), specialising in multimodal and graph learning for computational oncology. My Honours research develops a panel-agnostic survival model integrating spatial single-cell imaging, clinicopathologic variables and gene-level data, with patient-level leakage control and cross-cohort transfer. I am co-first author of an accepted Springer LNAI paper, co-author of two published peer-reviewed conference papers, and first author of two manuscripts under review. My PhD interests lie in multimodal biomedical AI, spatial omics and robust clinical translation.

EDUCATION

FEB 2023 – DEC 2026	Bachelor of Advanced Computing (Honours), Computational Data Science The University of Sydney , Sydney, Australia. <i>Thesis: “Hierarchical Niche-Aware Multimodal Learning for Breast Cancer Survival Prediction Using Imaging Mass Cytometry, Clinicopathologic Variables, and Gene-Level Data”, supervised by Associate Professor Xiuying Wang, School of Computer Science.</i> Cumulative WAM 82/100 (Distinction); First Class Honours expected, mark and class released December 2026.
2019 – 2022	High School Diploma, Specialised Physics Programme HUS High School for Gifted Students (HSGS) , VNU University of Science, Vietnam National University, Hanoi, Vietnam. GPA 9.5/10; outstanding academic achievement award, academic years 2019–2022.

RESEARCH INTERESTS

Biomedical AI Graph Learning Multimodal Learning Spatial Omics
Computational Oncology Machine Learning Deep Learning

PUBLICATIONS

† Equal contribution. Applicant’s name in bold. Quartiles: SCImago (SJR) 2025; ranks: CORE.

Journal Publications

- [J1] **B. M. Tran**[†], V. A. Nguyen[†], L. H. Nguyen, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, T. B. V. Nguyen, and D. H. Tran, “Discrimination was retained but calibration was not: development and external evaluation of a multimodal deep-learning model for breast cancer receptor classification,” *BMC Medical Informatics and Decision Making* (SJR Q1), *under review*, 2026.
- [J2] **B. M. Tran**, D. Q.-A. Nguyen, M.-H. Ha, D. M. Pham, T. D. Nguyen, Q. B. Doan, D. T. Doan, M.-A. Nguyen, D.-T. Nguyen, and O. T.-C. Chen, “H2F-GCN: Skeleton-based Hypergraph Frequency

Coordinate Fusion Graph Convolution Network for Physical Rehabilitation Assessment,” *IEEE Access* (SJR Q1), *under review*, 2026.

- [J3] L. H. Nguyen, **B. M. Tran**, and D. H. Tran, “Dual-Stream Contrastive Graph Representation Learning for lncRNA–Disease Association Prediction,” invited journal extension of [C3], *Vietnam Journal of Computer Science* (SJR Q3; WoS ESCI), *under review*, 2026.
- [J4] M.-H. Ha, X.-H. Le, A.-T. Dang, D.-P. Dao, H.-M.-D. Vu, D.-T. Kim, **B. M. Tran**, and O. T.-C. Chen, “GRAU++: An Efficient Lightweight Ghost-Residual-Attention U-Net++ Architecture for Automated Angle of Progression Biometry,” *EAI Endorsed Transactions on AI and Robotics* (SJR Q1), *under review*, 2026.

Peer-Reviewed Conference Publications

- [C1] **B. M. Tran**[†], V. A. Nguyen[†], L. H. Nguyen, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, T. B. V. Nguyen, and D. H. Tran, “A Multimodal Deep Learning Framework for Breast Cancer Receptor Status Prediction,” in *Proc. 12th International Symposium on Integrated Uncertainty in Knowledge Modelling and Decision Making (IUKM 2026)*, Quy Nhon, Vietnam, Sep. 2026. Springer LNAI. *Accepted, to appear*.
- [C2] L. H. Nguyen, **B. M. Tran**, and D. H. Tran, “A Variational Graph Neural Network with Multi-view Integration for Predicting lncRNA–Disease Associations,” in *Recent Challenges in Intelligent Information and Database Systems (ACIIDS 2026)*, CORE B, Communications in Computer and Information Science, vol. 2964, Springer, pp. 355–369, 2026. *Published*. doi:10.1007/978-981-92-0068-9_24
- [C3] L. H. Nguyen, **B. M. Tran**, and D. H. Tran, “Dual-Stream Contrastive Graph Representation Learning for lncRNA–Disease Association Prediction,” in *Proc. FISU Joint Conference on Artificial Intelligence (FJCAI 2026)*, Can Tho, Vietnam, Mar. 2026, pp. 151–158. *Published*. ISBN 978-604-45-2586-0
- [C4] **B. M. Tran**, L. Zhang, and X. Wang, “HiNT-Surv: Hierarchical Niche-Aware Multi-Micro-environment Learning from Imaging Mass Cytometry for Survival Prediction,” in *IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM), IEEE Symposium Series on Computational Intelligence (SSCI) 2027*, *under review*.

Other Publications

- [O1] **B. M. Tran**, “Promoting the Role of Overseas Vietnamese Youth in Protecting the Party’s Ideological Foundation in the Digital Space” (in Vietnamese), *State Management Review*, Academy of Administration and Public Management, Hanoi, Vietnam, 17 Aug. 2026. *Published*. [Full text](#)

RESEARCH EXPERIENCE

FEB 2026 –
PRESENT

Honours Research Student, [School of Computer Science](#), The University of Sydney, Sydney, Australia. Supervisor: A/Prof Xiuying Wang.

- Designed HiNT-Surv, a hierarchical niche-aware model that fuses imaging mass cytometry (IMC) with clinicopathologic and gene-level variables for survival prediction; evaluated on 404 METABRIC and 285 Basel patients (834 tissue cores, 50- and 52-marker panels).
- Parameterised the encoder by antigen identity rather than fixed channel order, so a model trained on one antibody panel transfers to another; kept all preprocessing and label-free pretraining inside each outer fold’s training partition to prevent patient-level leakage.
- Reached Harrell’s C-index 0.740 ± 0.020 (METABRIC) and 0.734 ± 0.030 (Basel) under patient-level cross-validation – gains of 0.028 and 0.057 over the clinical Cox-PH baseline – with median-risk stratification separating Kaplan–Meier curves in both cohorts (two-sided log-rank $p < 10^{-4}$).
- Trained on METABRIC and evaluated on Basel without target-cohort fine-tuning, reaching C-index 0.697 against 0.631 for the strongest published comparator, with a smaller source-to-target decrease (0.043 vs 0.081).

DEC 2025 –
PRESENT

Research Collaborator, Department of Obstetrics & Gynaecology, [Hanoi Medical University](#), Hanoi, Vietnam. Multimodal breast-cancer receptor prediction.

- Co-developed a multimodal, multitask deep-learning framework integrating clinical variables, H&E whole-slide images and tumour RNA-seq for joint ER, PR and HER2 prediction; reported in [C1].
- Evaluated with patient-level stratified five-fold cross-validation on 937 TCGA-BRCA cases (macro AUC 0.890; ER 0.954, PR 0.868, HER2 0.848) and frozen external testing on 122 CPTAC-BRCA cases without retraining, tuning or threshold adjustment (macro AUC 0.931).

DEC 2025 –
MAR 2026

Research Assistant, Center for Biomedical and Pharmaceutical Research, [Vietnam Academy of Science and Technology \(VAST\)](#), Hanoi, Vietnam.

- Applied reproducible NGS preprocessing and annotation workflows (FastQC, Trimmomatic, genome assembly, gene prediction and functional annotation) to mutation and microbial gene-function analyses.

RESEARCH PROJECTS

DEC 2026 –
NOV 2027

Development and Calibration of an AI Model for Individualised Post-Test Probability after High-Risk NIPT (Trisomy 21, 18 and 13)

- Role: designing and calibrating the probability model – a retrospective cohort of high-risk NIPT referrals, converting a screen-positive label into an individualised probability of confirmed fetal aneuploidy from signed Z-score, fetal fraction and assay platform.
- Principal investigator: Nguyen Van An; supervisor: Nguyen Thi Bich Van, MD, PhD, Department of Obstetrics & Gynaecology, [Hanoi Medical University](#). Institutional research scheme, 12 months, total budget 10,000,000 VND – *proposal submitted*.

AWARDS & SCHOLARSHIPS

2026

Outstanding Student Award, [Embassy of Vietnam in Australia](#) – for academic and research achievement.

2026

Second Prize, Vietnamese Young Intellectuals Forum in Australia – for the multimodal breast-cancer receptor-status research [C1].

2026

Third Prize, Youth Science and Technology Conference, [Hanoi Medical University](#) – for [C1]; work selected for the national health-sector conference.

2025

January Star Award, [Vietnam National Union of Students](#) – one of 117 national recipients, for academic achievement, research and student leadership.

2023 –
2026

Sydney International Student Award, The University of Sydney – merit-based tuition scholarship awarded at admission and held for the full degree.

INDUSTRY EXPERIENCE

DEC 2024 –
MAR 2025

Data Engineer Intern, [Viettel Software](#), Hanoi, Vietnam.

- Built and maintained Python/SQL ETL pipelines for operational datasets; contributed to data-quality monitoring, schema design and query/pipeline performance tuning supporting analytics teams.

JUN 2024 –
JUL 2024

Data Engineer Intern, [Agribank](#) (Vietnam Bank for Agriculture and Rural Development), Hanoi, Vietnam.

- Supported ETL of banking datasets into reporting systems; contributed to dashboards and operational reports used by internal stakeholders.

TECHNICAL SKILLS

PROGRAMMING	Python, R, SQL, Bash.
DEEP LEARNING	PyTorch, PyTorch Geometric, scikit-learn.
METHODS	Graph neural networks, self-supervised and contrastive learning, multimodal fusion, survival modelling.
EVALUATION	Patient-level stratified cross-validation, bootstrap confidence intervals, paired hypothesis testing, multiple-testing correction, effect-size analysis, external validation.
BIOMEDICAL DATA	Spatial single-cell IMC, RNA-seq / gene expression, histopathology whole-slide images.
RESEARCH COMPUTING	Git, Linux, experiment tracking.
LANGUAGES	Vietnamese (native); English (fluent – all tertiary study in English).

LEADERSHIP & SERVICE

MAR 2026 – PRESENT	Chief of Administrative Office, Vietnamese Students in Australia (SVAU). • Manage administrative operations, advisory functions and student science and technology activities.
MAR 2026 – PRESENT	Executive Committee Member, Vietnam–Australia Innovation Network (NIC-AU). • Support network governance and initiatives linking Vietnamese researchers in Australia with the Vietnam innovation ecosystem.

REFERENCES

REFeree 1	Associate Professor Xiuying Wang School of Computer Science, The University of Sydney, Australia. Email: xiu.wang@sydney.edu.au
REFeree 2	Associate Professor Dang Hung Tran Hanoi University of Industry , Hanoi, Vietnam. Email: hungtd@haui.edu.vn
REFeree 3	Dr Manh Hung Ha Faculty of Applied Sciences, International School, Vietnam National University, Hanoi (VNU-IS) . Email: hungmh@vnu.edu.vn